Deven Desai

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SUMMARY

Software Engineering Master's student and Stanford Hackathon winner with hands on experience building production ML and generative AI systems, RAG pipelines, and automated model evaluation loops.

PROJECTS

OmniTrack AI | LLM Evaluation & Document Processing Platform GitHub

- Processed unstructured streaming document inputs and metadata natively in cloud infrastructure by engineering a zero-to-one platform using Python and React.
- Achieved sub-100ms vector search latency by independently optimizing Qdrant retrieval algorithms for compute-intensive document queries.
- Enhanced system correctness and output reliability by 30 percent by building continuous LLM evaluation loops and automated verification test suites.

AgenticFlow | Generative AI Workflow & Execution Engine GitHub

- Automated enterprise schema migrations and complex data transformations across cloud environments by architecting an end-to-end execution engine using Python, Go, and LangGraph.
- Increased service execution throughput by 40 percent by designing concurrent reasoning loops and gRPC microservice interfaces as measured by backend traces.
- Ensured high system reliability and data privacy across platform boundaries by implementing an automated differential privacy validation layer.

High-Performance Search Engine | Information Retrieval Discovery Interface GitHub

- Analyzed and queried high-volume structured datasets in real time by engineering a custom full-stack analytical search interface as a solo developer.
- Decreased search query latency by 35 percent by developing optimized retrieval algorithms using Python and Java as measured by end-to-end response times.
- Optimized frontend-backend data execution throughput by 20 percent by tuning underlying data structures and batch processing routines.

WORK EXPERIENCE

Applied AI Intern, *Parspec, Inc., USA* **May 2026 – Dec 2026**

- Improved product-search relevance over a distributor inventory catalog, raising **recall@3 from 50% to 90%** on a **5,000-query** labeled evaluation set, by replacing keyword-only search with **hybrid** lexical and **semantic retrieval** fused via **reciprocal rank fusion**, re-scored with a **cross-encoder re-ranker**.
- Eliminated manual review of several-hundred-page construction documents for distributor bids, extracting structured product records at **88%** precision across **CSI Divisions 26 and 22**, by building a **RAG** pipeline over written specifications and schedules that grounds every extracted field in its source excerpt and auto-enriches contractor **Quantity Take-Off** line items.
- Automated lighting fixture product substitutions across 1.2M+ indexed datasheets at 85 percent match precision by engineering a retrieval pipeline that resolves free text inputs to anchor products and scores candidates via multi attribute filtering and semantic search.

Full Stack Developer, *Terminal Trends, India* **May 2024 – Dec 2024**

- Developed and optimized RESTful APIs using **Node.js** and **Express** to integrate real-time EV charging station data from multiple vendors (e.g., Tesla, government APIs), reducing server response time by **40%** and improving data reliability across the platform.
- Built interactive route-planning features with **React** and **Redux**, enabling users to locate nearby EV stations, view waiting times, pricing per kWh, and other live metrics along their journey, enhancing user convenience and engagement.

TECHNICAL SKILLS

AI & Machine Learning: Generative AI Systems, Retrieval-Augmented Generation (RAG), Agentic Workflows (LangGraph), Model Evaluation Loops, Vector Search (Qdrant), Cross-Encoders, PyTorch, Hugging Face
Languages: Python, TypeScript, JavaScript, SQL, C++, Go, Java
Full Stack & APIs: React, Redux, Node.js, Express, RESTful APIs, gRPC Microservices, Relational Schema Design
Infrastructure & Tools: Docker, CI/CD, Git, GitHub, Production Monitoring, Automated Unit Testing, System Design

EDUCATION

San Jose State University **San Jose, CA**
Master of Science in Software Engineering

ACHIEVEMENTS

- **Winner – Davis Wright Tremaine LLP Prize**, *Stanford Codex LLM x Law Hackathon (2025)*
  Led a 4-member team to build a **RAG-based legal QA engine** using Python, GPT-4, and vector-embedding retrieval with **cosine similarity**, enabling accurate extraction of contract clauses and delivering **90%+ correct answers**, outperforming 30+ teams.